Given: Due:	Further Maths (Decision) HW 10	Name:

Chapter 3: Algorithms on graphs (MST and shortest paths)

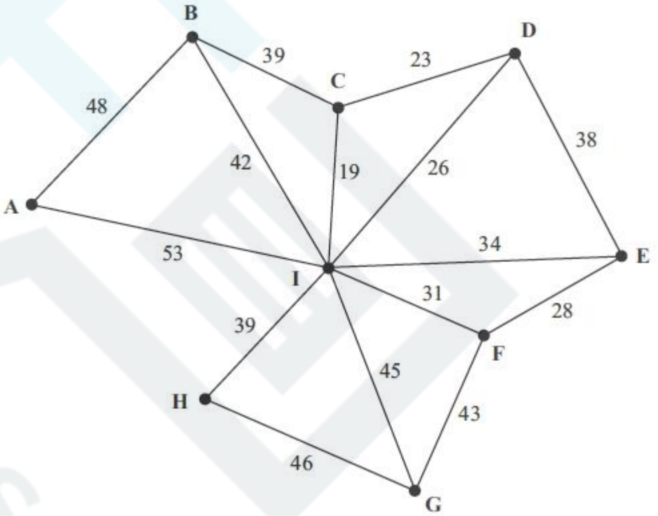
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)

Useful resources: Your notes from recent lessons & your text book

Questions - do these on lined paper

1) (a) Use Kruskal's algorithm to find a minimum spanning tree for the network shown in Figure 2. You should list the arcs in the order in which you consider them. In each case, state whether you are adding the arc to your minimum spanning tree. (3)



(b) Starting at A, use Prim's algorithm to find a minimum spanning tree for the network in Figure 2. You must clearly state the order in which you include the arcs in your tree. (3)

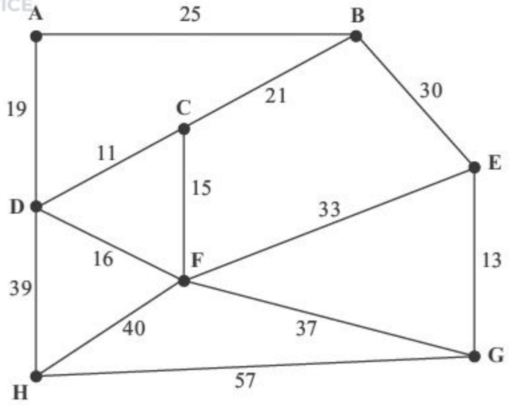
(c) Draw a minimum spanning tree for the network in Figure 2 using the vertices given in Diagram 1 of the answer book. State the weight of the minimum spanning tree. (2)

A new spanning tree is required which includes the arcs DI and HG, and which has the lowest possible total weight.

(d) Explain which algorithm you would choose to complete the tree, and how the algorithm should be adapted. (You do not need to find the tree.)

(2)

2) Figure 1 represents the distances, in km, between eight vertices, A, B, C, D, E, F, G and H in a network.



(a) Use **Kruskal's** algorithm to find the minimum spanning tree for the network. You should list the arcs in the order in which you consider them. In each case, state whether you are adding the arc to your minimum spanning tree.

(3)

(b) Starting at A, use **Prim's** algorithm to find the minimum spanning tree. You must clearly state the order in which you
selected the arcs of your tree.

(3)

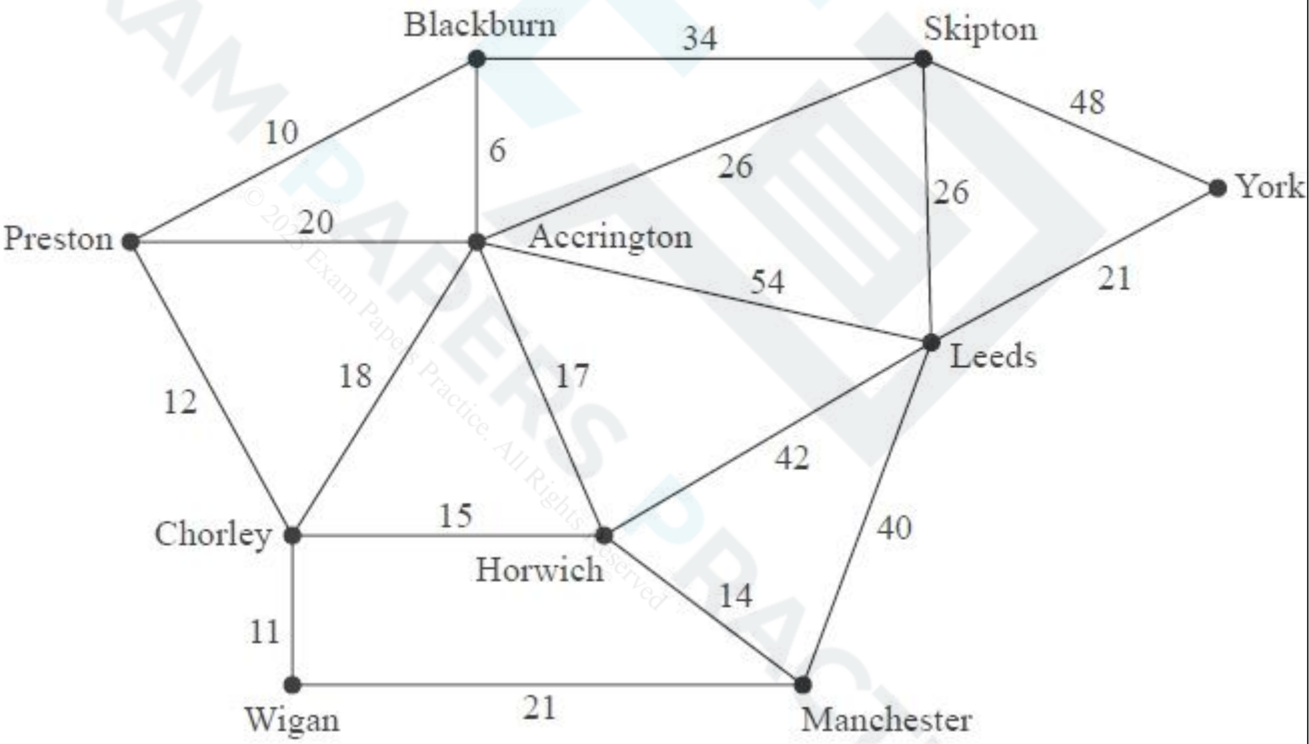
(c) Draw the minimum spanning tree using the vertices given in Diagram 2 in the answer book.

(1)

(d) State the weight of the tree.

(1)

3)



Sharon is planning a road trip from Preston to York. Figure 2 shows the network of roads that she could take on her trip. The number on each arc is the length of the corresponding road in miles.

(a) Use Dijkstra's algorithm to find the shortest route from Preston (P) to York (Y). State the shortest route and its length.

(6)

Sharon has a friend, John, who lives in Manchester (M). Sharon decides to travel from Preston to York via Manchester so she can visit John. She wishes to minimise the length of her route.

(b) State the new shortest route. Hence calculate the additional distance she must travel to visit John on this trip. You must make clear the numbers you use in your calculation.

(3)

END OF HOMEWORK
Total = 27 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.
File this question paper in your file, ready for revision and file inspection.

Answer book

1)

(c)

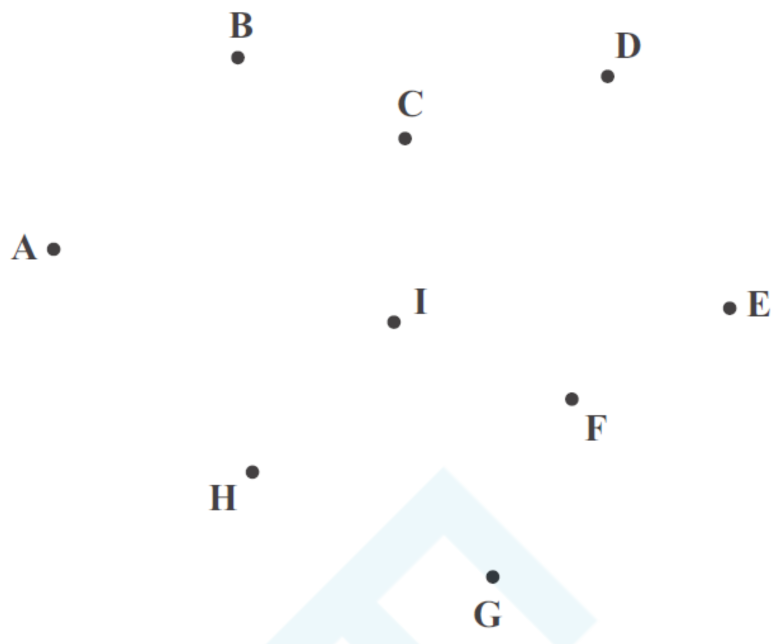


Diagram 1

Weight of minimum spanning tree: ________________________________

2)

(c)

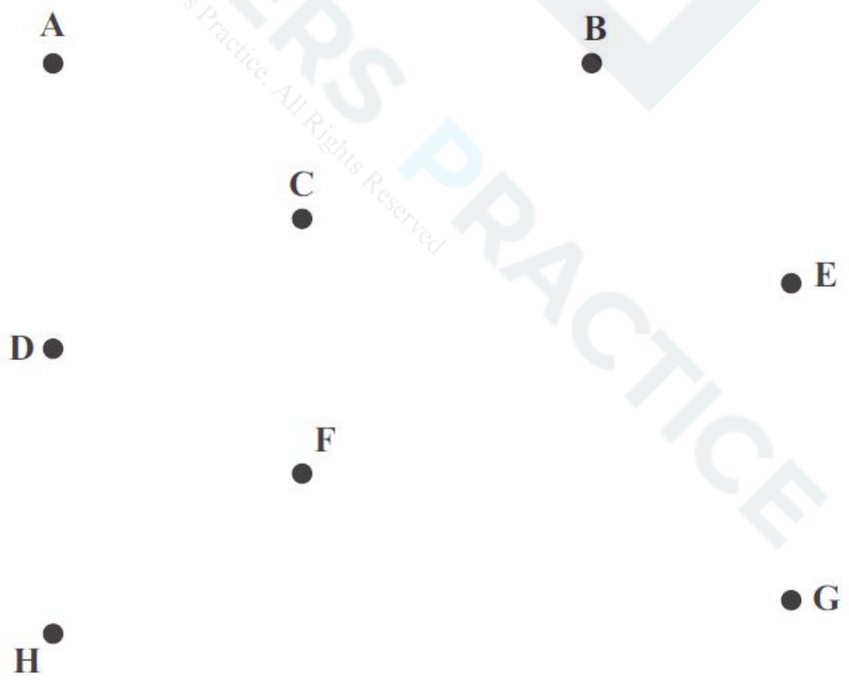
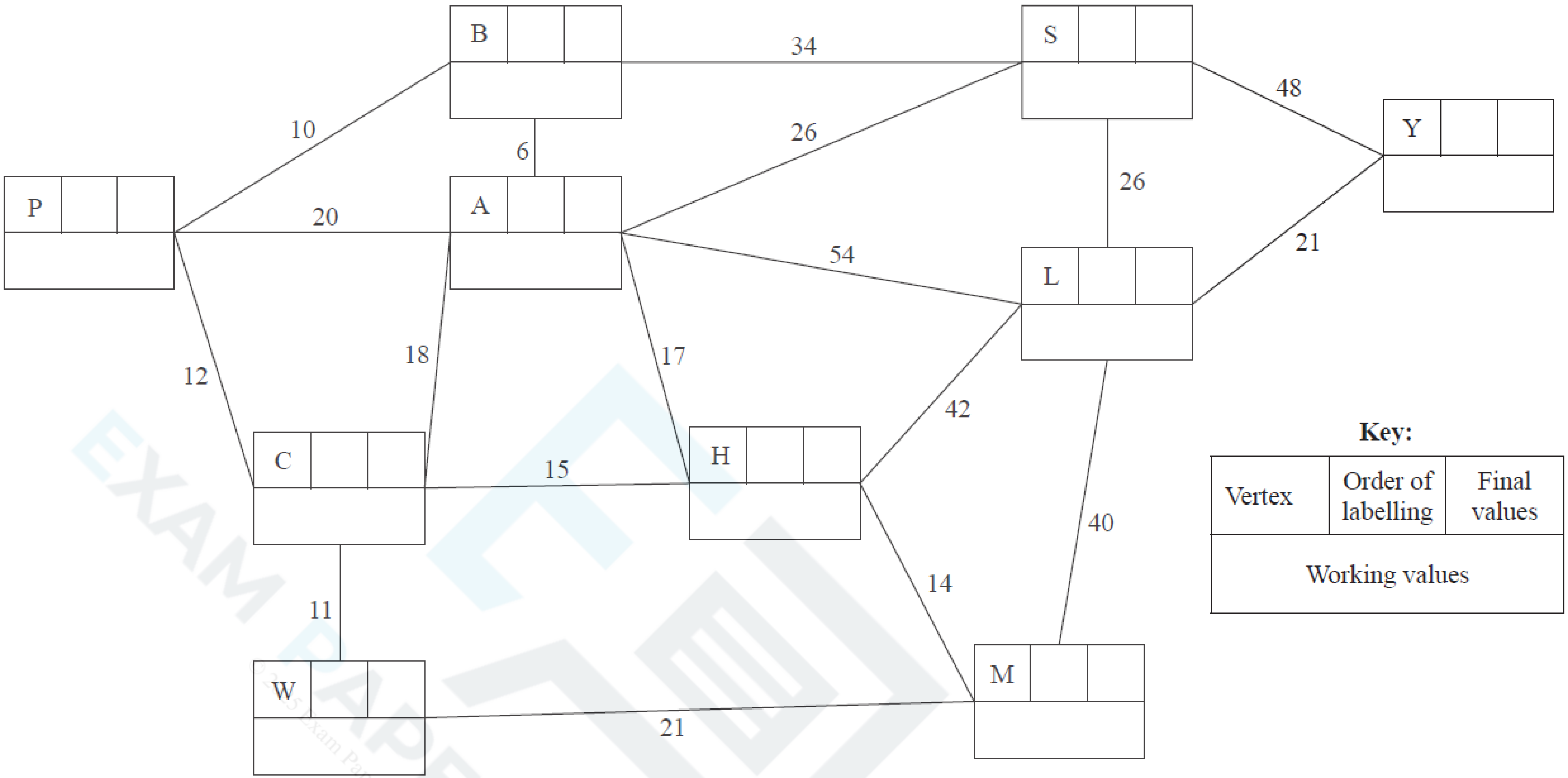


Diagram 1

(d) Weight of tree = ...

3



Shortest route: ______________________________

Length of shortest route: ______________________________

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